

A – 3949

MCA (First Semester) EXAMINATION,
Dec.-2023

Paper-MCA-104

COMPUTER ORGANIZATION AND
PROGRAMMING LANGUAGE
PROGRAMMING

Time : Three Hours

Maximum Marks : 60

Minimum Pass Marks : 21

Note—Attempt *all* questions. Question
No. 1 is compulsory.

1. Attempt any *five* questions (Answers not
more than 50 words)—

- (i) What is Binary number ? How it is
represented ? 2
- (ii) Define octal system. 2

P.T.O.

(2)

A - 3949

- (iii) What is De-multiplexers? 2
- (iv) What is the function of Control Unit? 2
- (v) Define Address Bus and Data Bus. 2
- (vi) What do you mean by data processing? 2
- (vii) What is interrupt? How many types of interrupts exists in computer system? 2
- (viii) What is logical unit? 2
- (ix) Define secondary memory. 2
- (x) Give three difference between Dynamic RAM and Static RAM. 2

Unit-I

2. How do computers use different number systems to represent information? Explain the various types of number systems. 10

Or

Explain error detection and correction codes. 10

(3)

A - 3949

Unit-II

3. What is logic gates? Explain the various logic gates with truth tables. 10

Or

What is Registers? What is the role of registers in computer system? Discuss the various types of registers used in computer system. 10

Unit-III

4. Explain the Register Transfer Language with the help of diagram. 10

Or

What is addressing mode? Explain the various types of addressing modes. 10

Unit-IV

5. Explain the conditional and unconditional looping in detailed. 10

Or

Explain I/O organization in detailed. 10

P.T.O.

Unit-V

6. Write a program in 8086/8088 assembly language to add of two eight bytes data. 10

Or

Explain associative memory and set associative memory. 10